

Coding & Solution

Date	8 July 2026
Team ID	SWTID-2026-8977
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Rakesh1115/EduGenie-Google-Powered-Learning-Assistant
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	FastAPI, Google Gemini API

Key Features Implemented	• Intelligent Question Answering using Google Gemini • Simplified Concept Explanation • AI-Powered Quiz Generation • Educational Text Summarization • Personalized Learning Path Recommendations • Responsive Web Interface using HTML, CSS, and JavaScript • FastAPI Backend for handling user requests • Google Gemini API integration for AI-generated educational responses
Pending / Incomplete Features	• User Authentication and Login System (Future Enhancement) • User Profile and Learning Progress Tracking • Quiz Score Analytics Dashboard • Multi-language Learning Support • Voice-based Question Input
Setup / Run Instructions	1. Clone the GitHub repository. 2. Create and activate a Python virtual environment. 3. Install dependencies using <code>pip install -r requirements.txt</code>. 4. Add your Google Gemini API Key to the <code>.env</code> file. 5. Start the FastAPI server using <code>uvicorn main:app --reload</code>. 6. Open the frontend in a web browser and interact with EduGenie.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes

4	Error handling is implemented for critical operations	Yes
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5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- ☐ The project follows a modular architecture with separate frontend and backend components.
- ☐ FastAPI is used to build a lightweight and high-performance REST API.
- ☐ Google Gemini API powers intelligent educational assistance, including question answering, summarization, quiz generation, and learning path recommendations.
- ☐ HTML, CSS, and JavaScript provide a clean and responsive user interface.
- ☐ The code follows Python coding standards (PEP 8), uses meaningful naming conventions, and includes reusable functions for better maintainability.
- ☐ The application is designed to be lightweight, scalable, and suitable for educational use on resource-constrained devices.